

Barycentric Coordinates: Trading a Point of Origin for a Triangle

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1 Introduction

Mathematics is, in many ways, a challenge of finding the right perspective from which to view a problem. In combinatorics, this may translate into counting objects in a certain order, while number theorists might interpret it as finding a useful factorization of an algebraic expression, for instance. Geometry is full of such examples. There are subtle instances that simply involve changing the way you look at the interactions between the parts of a diagram; one of my favorites, despite its simplicity, is translating the power of a point to the language of similar triangles.

Then, there are more drastic examples, such as assigning an ordered pair to each point in the plane, as in Cartesian coordinates or polar coordinates, or assigning a unique complex number to each point, as in the complex plane. Each coordinate system has its own specialties: Cartesian coordinates are well equipped for understanding configurations involving numerous lines and their intersections, while polar coordinates make natural the language of rotation around some central point. Complex numbers are particularly interesting, for they are a strange amalgam of the previous two systems.

In this article, we explore another one of these coordinate systems: barycentric coordinates, which are specially equipped to work with ratios between areas and lengths relative to a central triangle.

2 A Brief Primer of Barycentric Coordinates

Barycentric coordinates are centered around a *reference triangle*, which we will refer to as $\triangle ABC$ throughout this article. Essentially, they fix the coordinates of the vertices of a specific triangle, and build around them such that all other vertices are expressed in terms of this central triangle.

In 3D space, let $\triangle ABC$ be positioned such that no three of points O , A , B , and C are collinear, where O is the origin of the barycentric plane. Then, any vector P whose endpoint is in the plane of $\triangle ABC$ can be expressed as

$$\vec{P} = \alpha \vec{A} + \beta \vec{B} + \gamma \vec{C}$$

with $\alpha + \beta + \gamma = 1$. This ordered triple (α, β, γ) is defined as the coordinates of P .

2.1 Coordinates as Area Ratios

Geometrically, we can interpret the coordinates as area ratios, as shown in the following theorem.

2.1.1 Theorem

The barycentric coordinates of a point P are

$$\left(\frac{[PBC]}{[ABC]}, \frac{[PCA]}{[ABC]}, \frac{[PAB]}{[ABC]} \right),$$

where $[\lambda]$ denotes the signed area¹ of the geometric figure λ , and where we take P with ratio to the reference triangle ABC .

2.1.2 Proof

We can apply a linear transformation such that $\vec{A}$, $\vec{B}$, and $\vec{C}$ are the vectors $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ respectively. Then, let P have barycentric coordinates (α, β, γ) ; $\vec{P}$ is by definition (α, β, γ) . This means the determinant

$$\begin{vmatrix} \alpha & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \alpha$$

gives the ratio of the volumes of tetrahedra $OPBC$ and $OABC$, where O is the origin. Since the bases of the tetrahedra are coplanar, the tetrahedra's heights are equivalent, so this ratio is equivalent to the ratio of the signed areas of $\triangle PBC$ and $\triangle ABC$, as claimed by the statement of the theorem. An analogous argument tells us that β and γ are $\frac{[PCA]}{[ABC]}$ and $\frac{[PAB]}{[ABC]}$ respectively.

2.1.3 Implications

As a result of the previous theorem, the following characteristics of barycentric coordinates are made apparent.

1. The coordinates of P are unique; that is, if two points P_1 and P_2 both have the same coordinates, then they are the same point.

¹Taking ABC to always have positive area, if λ is oriented the same way as ABC , then its signed area is its absolute area (i.e., it is positive); otherwise, the signed area is the negative of its absolute area.

2. P lies inside $\triangle ABC$ if and only if α , β , and γ are all positive.
3. P lies on an edge of $\triangle ABC$ if and only if at least one of the coordinates is 0.
4. P is outside $\triangle ABC$ if and only if at least one of α , β , γ is negative.
5. A , B , and C have coordinates $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$, respectively.

While the coordinates in their current form are useful in their own right, it is sometimes helpful to entirely get rid of the constraint $\alpha + \beta + \gamma = 1$: given an arbitrary triple of numbers (α, β, γ) with $\alpha + \beta + \gamma \neq 0$, we can divide each number by $\alpha + \beta + \gamma$ to get a triple of numbers whose sum is one. In particular, this means we can simply consider the ratio of $\alpha : \beta : \gamma$ to be the point P . When the coordinates are written this way, we use the notation $(\alpha : \beta : \gamma)$ to indicate that we are only looking at the ratio of the coordinates. Coordinates of this form are called *dehomogenized coordinates*.

An application of these is shown in the following theorem:

2.1.4 Theorem

Let D be a point on side BC of $\triangle ABC$ such that $BD : DC = d : 1 - d$.² Then, the set of points on line AD (excluding A) are of the form $(t : 1 - d : d)$, where t is a real number.

2.1.5 Proof

Let P be a point on line AD with $P \neq A$. Then, since $BD : DC = d : 1 - d$, the ratio of $[PCA] : [PAB]$ is $1 - d : d$. The area of $\triangle PBC$ is entirely arbitrary here, so the coordinates of P are $(t : 1 - d : d)$ where t can take on any real value. This fails to include the coordinates of A because $[PCA]$ and $[PAB]$ have zero area when $P = A$, so $t : 1 - d : d = 1 : 0 : 0$, which is possible only if $1 - d = d = 0$, but this is itself impossible.

In particular, dehomogenized coordinates can help us determine the characteristics of collinearity, as seen above.

2.1.6 Exercise

Prove Ceva's theorem; namely, given points D , E , F on sides BC , CA , AB of $\triangle ABC$, prove that AD , BE , and CF concur if and only if

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1,$$

where lengths are directed.

2.1.7 Exercise

What are the coordinates of the incenter, centroid, circumcenter, and orthocenter in terms of the side lengths $a = BC$, $b = CA$, and $c = AB$ of ABC ?

²Here, lengths are directed, meaning that given collinear points A , B , C , $AB + BC = AC$ is true regardless of the permutation of the points. Specifically, lengths are negated as necessary to satisfy this relation.

2.2 Properties of Lines

Let $\triangle PQR$ be a triangle that lies in the plane of $\triangle ABC$, with P, Q, R having barycentric coordinates $(p_1, p_2, p_3), (q_1, q_2, q_3), (r_1, r_2, r_3)$. Then, we can determine ratio of the signed areas of $\triangle PQR$ and $\triangle ABC$ with the determinant

$$\begin{vmatrix} p_1 & p_2 & p_3 \\ q_1 & q_2 & q_3 \\ r_1 & r_2 & r_3 \end{vmatrix}.$$

P, Q , and R are collinear if and only if the area of $\triangle PQR$ is 0, which means they are collinear if and only if the aforementioned determinant is 0. This allows us to obtain a general equation of a line.

2.2.1 Theorem

The equation of a line in barycentric coordinates is of the form

$$ux + vy + wz = 0,$$

where (x, y, z) is some point on the line and u, v, w are some constants.

2.2.2 Proof

Let $S = (s_1 : s_2 : s_3)$ and $T = (t_1 : t_2 : t_3)$ be arbitrary points in the plane of $\triangle ABC$. Then, any point $P = (x, y, z)$ on line ST satisfies

$$\begin{vmatrix} x & y & z \\ s_1 & s_2 & s_3 \\ t_1 & t_2 & t_3 \end{vmatrix} = 0.$$

Evaluating this determinant by minors yields

$$(s_2 t_3 - t_2 s_3)x + (s_3 t_1 - t_3 s_1)y + (s_1 t_2 - t_1 s_2)z = 0.$$

Then, define the constants u, v , and w to be the coefficients of x, y , and z respectively, which tells us that the set of points on a line is defined by the equation

$$ux + vy + wz = 0$$

as desired.

2.2.3 Implications

The power of this relatively simple form of a line is that it is easy to find where lines intersect: the intersection points come from solutions to the system

$$\begin{aligned} u_1 x + v_1 y + w_1 z &= 0 \\ u_2 x + v_2 y + w_2 z &= 0 \\ x + y + z &= 1. \end{aligned}$$

If the system has no solutions, then the lines must of course be parallel. However, there is something more subtle here; since the system has no solutions, we have

$$\begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ 1 & 1 & 1 \end{vmatrix} = 0.$$

But this implies that

$$\begin{aligned} u_1x + v_1y + w_1z &= 0 \\ u_2x + v_2y + w_2z &= 0 \\ x + y + z &= 0 \end{aligned}$$

has a nontrivial solution!

This apparently nonsensical triple of coordinates allows for the introduction of the concept of *points at infinity*. The point at infinity of any line is the nontrivial triple of coordinates satisfying the equation of the line whose coordinates sum to zero; this now means that every pair of lines intersects at a unique point, including parallel lines. This makes the following theorem more powerful, as we now have a way to interpret coordinates whose sum is zero.

2.2.4 Theorem

Three lines $u_ix + v_iy + w_iz = 0$ for $i = 1, 2, 3$ are either concurrent or all parallel if and only if

$$\begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{vmatrix} = 0.$$

2.2.5 Proof

To find the intersection point of the three lines, we have a system of equations in x, y, z :

$$\begin{aligned} u_1x + v_1y + w_1z &= 0 \\ u_2x + v_2y + w_2z &= 0 \\ u_3x + v_3y + w_3z &= 0 \end{aligned}$$

Obviously, the system has the trivial solution $(x, y, z) = (0, 0, 0)$, but it has nontrivial solutions if and only if the matrix representing this system of equations has determinant zero; that is, the existence of a common intersection point on the three lines is equivalent to

$$\begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{vmatrix} = 0.$$

Now, supposing the determinant is indeed zero, solutions to this system must satisfy either $x + y + z = 0$ or $x + y + z \neq 0$. If $x + y + z \neq 0$, then the lines intersect at a point $(x : y : z)$ in the plane, meaning that they concur. On the other hand, if $x + y + z = 0$, the lines intersect at a point at infinity, so they are all parallel.

3 Applications of Barycentric Coordinates

Now that we are familiar with the most essential properties of barycentric coordinates, it helps to look at some applications to see how they relate to various areas of math.

3.1 Olympiad Geometry

One such application is their use in math olympiads. To demonstrate this, let's look at a problem from the 2016 Indian National Mathematical Olympiad:

3.1.1 Problem

(INMO 2016/1) Let ABC be a triangle in which $AB = AC$. Suppose the orthocenter of the triangle lies on the incircle. Find the ratio AB/BC .

3.1.2 Solution

First, what suggests that this problem can be solved with barycentric coordinates? The fact that we're asked to find a ratio of side lengths of the triangle is a good indicator, since the triangle's side lengths appear in the coordinates of both the incenter and the orthocenter.

However, what about the circle? We don't know how to deal with circles! Fortunately, the fact that $\triangle ABC$ is isosceles implies that the incenter I lies on the altitude from A to BC , and the point where the incircle touches BC is the foot of that altitude; we will call that point M . Thus, we can deduce that the foot of the altitude from A is diametrically opposite the orthocenter, which we will label H . In other words, I is the midpoint of HM ; this relation allows us to entirely ignore the circle for the rest of the problem!

Additionally, some useful notation before continuing: define

$$S_A = \frac{b^2 + c^2 - a^2}{2} = ab \cos C,$$

where a, b, c are the lengths of the sides of $\triangle ABC$ opposite A, B, C respectively, and define S_B and S_C analogously. This relation allows us to simplify otherwise messy expressions, such as that of the orthocenter, which we can write as

$$H = (S_B S_C : S_C S_A : S_A S_B).$$

Now, continuing the problem: since we are only interested in the ratio of $AB : BC$, let $BC = 1$ and $AB = AC = a$, so that a is the ratio we want to find. I has coordinates $(1 : a : a)$ and M , being the midpoint of BC , has coordinates $(0 : 1 : 1)$. We also get that

$$S_A = a^2 - \frac{1}{2} \text{ and } S_B = S_C = \frac{1}{2},$$

so the coordinates of H are

$$(1 : 2a^2 - 1 : 2a^2 - 1).$$

However, we can also write the coordinates of H using the incenter. Since IM is half of HM , the area of $\triangle BHC$ is twice that of $\triangle IHC$. Hence,

$$\frac{[HBC]}{[ABC]} = \frac{2[IBC]}{[ABC]} = \frac{2}{2a + 1}.$$

By the symmetry in the isosceles triangle,

$$[HCA] = [HAB],$$

which implies that

$$\frac{[HAB] + [HBC] + [HCA]}{[ABC]} = \frac{2}{2a+1} + \frac{2[HCA]}{[ABC]} = 1.$$

Therefore,

$$\frac{[HCA]}{[ABC]} = \frac{[HAB]}{[ABC]} = a - \frac{1}{2},$$

meaning $H = (4 : 2a - 1 : 2a - 1) = (1 : 2a^2 - 1 : 2a^2 - 1)$. Since these triples of coordinates designate the same point,

$$\frac{4}{2a-1} = \frac{1}{2a^2-1}.$$

This yields the quadratic equation $8a^2 - 2a - 3 = 0$, whose only positive solution is $a = \boxed{\frac{3}{4}}$.

In general, olympiad geometry problems that involve lots of lines and ratios of areas or segments are likely to be solvable with barycentric coordinates, while problems involving circles are typically tough to work with.

3.2 Computer Graphics

Instead of defining barycentric coordinates on a plane in three-dimensional space, it is possible to define them in two dimensions. This makes it easier to convert between barycentric coordinates and two-dimensional systems, which computers can more readily interpret due to their screens being rectangles. Each point on the computer screen can then be assigned barycentric coordinates relative to some triangle. The coordinates of points inside the triangle (which are nicely filtered out by the signs of the coordinates) can be converted to a value between 0 and 255, so that each point is a triple of numbers from 0 to 255, which gives it an RGB color. This results in a smooth gradient between red, green, and blue.

There have also been attempts to generalize barycentric coordinates to polygons with more vertices, some of which have been used for color interpolation, image warping, and shape deformation. However, a closed-form construction of barycentric coordinates has still not been found.

4 Connections to Other Coordinate Systems

Barycentric coordinates are closely related to trilinear coordinates, which define points by signed distance to the edges of the triangle. For some point P and $\triangle ABC$, the area of $\triangle PAB$ is half the distance from P to AB times the length of AB , so a triple of trilinear coordinates $(x : y : z)$ is equivalent to $(ax : by : cz)$ in barycentric coordinates.

Using our three-dimensional formulation of barycentric coordinates, some linear algebra can be used to convert barycentrics to regular three-dimensional coordinates. The two-dimensional definition that was previously mentioned can similarly be used to convert them to Cartesian coordinates as well.

References

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